

Study questions for regression models:

Chapter 7

Section 7.1

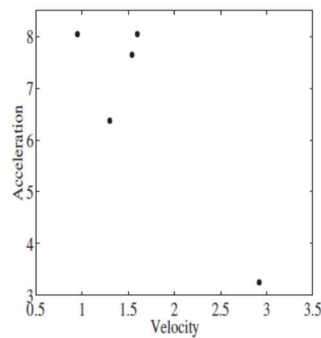
4. (a) True. This is a result of the plot being clustered around a line with positive slope.
- (b) False. Since the plot is clustered around a line with negative slope, below average values of one variable will tend to be associated with above average values of the other.
- (c) False. The correlation coefficient describes the shape of the plot, not its location relative to the axes.

6. (a) Let x represent velocity and let y represent acceleration.

$$\bar{x} = 1.662, \quad \bar{y} = 6.668, \quad \sum_{i=1}^n (x_i - \bar{x})^2 = 2.23928, \quad \sum_{i=1}^n (y_i - \bar{y})^2 = 16.48108, \\ \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = -5.37248.$$

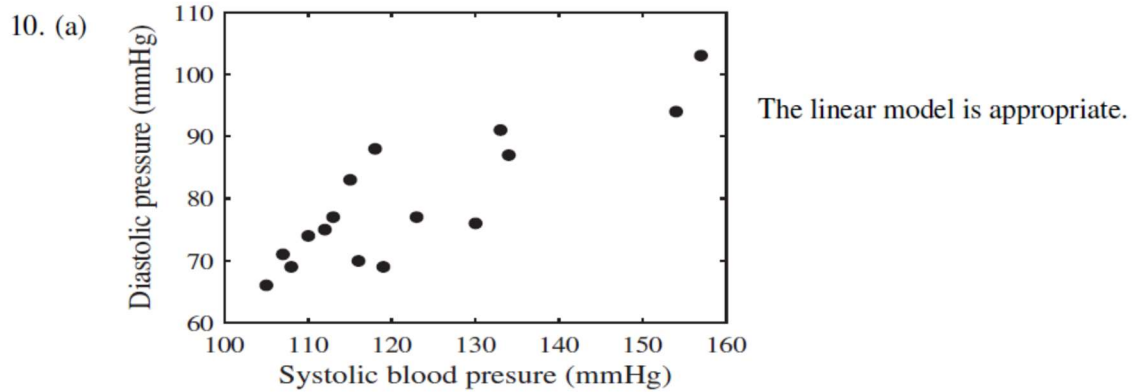
$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = -0.88436.$$

(b)



- (c) No, the point (2.92, 3.25) is an outlier.
- (d) No effect. Converting units from meters to centimeters and from seconds to minutes involves multiplying by a constant, which does not change the correlation coefficient.

Section 7.2



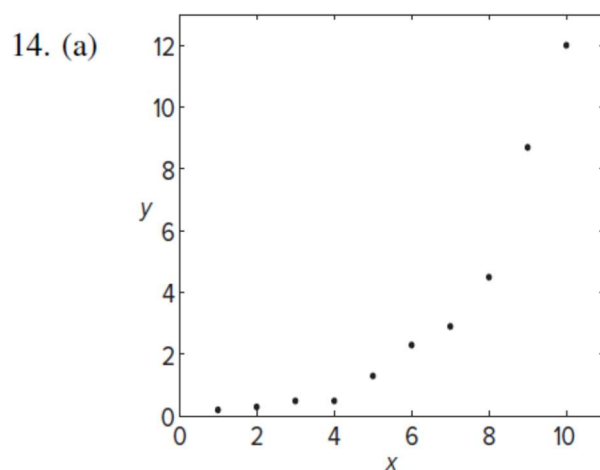
(b) $\bar{x} = 122.125$, $\bar{y} = 79.375$, $\sum_{i=1}^n (x_i - \bar{x})^2 = 3723.75$, $\sum_{i=1}^n (y_i - \bar{y})^2 = 1675.75$,
 $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 2140.25$.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = 9.1828466 \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.574757.$$

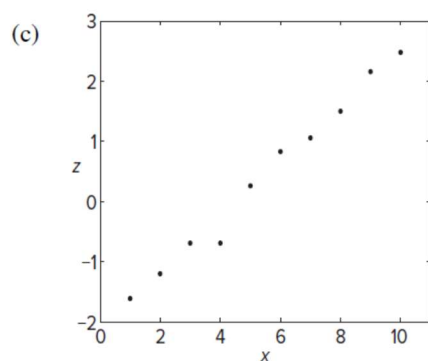
The equation of the least-squares line is $y = 9.1828466 + 0.574757x$.

(c) By $0.574757(10) = 5.74757$ mmHg.

(d) $9.1828466 + 0.574757(125) = 81.03$ mmHg.



(b) No, because x and y are not linearly related.



(d) $\bar{x} = 5.5$, $\bar{z} = 0.41$, $\sum_{i=1}^n (x_i - \bar{x})^2 = 82.5$, $\sum_{i=1}^n (z_i - \bar{z})^2 = 18.2494$, $\sum_{i=1}^n (x_i - \bar{x})(z_i - \bar{z}) = 38.55$.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(z_i - \bar{z})}{\sum_{i=1}^n (x_i - \bar{x})^2} = 0.4672727 \text{ and } \hat{\beta}_0 = \bar{z} - \hat{\beta}_1 \bar{x} = -2.16.$$

The equation of the least-squares line is $z = -2.16 + 0.4672727x$.

The predicted value of z when $x = 4.5$ is $\hat{z} = -2.16 + 0.4672727(4.5) = -0.057273$.

This is an appropriate method of prediction, because x and z are linearly related.

(e) $\hat{y} = e^{\hat{z}} = e^{-0.057273} = 0.944$. Since $\hat{z}$ is a reasonable predictor of z , and since $y = e^z$, $e^{\hat{z}}$ is a reasonable predictor of y .

Section 7.3

2. (a) $n - 2 = 7 - 2 = 5$

(b) $\hat{\beta}_1 = -0.27985$, $s_{\hat{\beta}_1} = 0.079258$, $t_{5,01} = 3.365$, so a 98% confidence interval is $-0.27985 \pm 3.365(0.079258)$, or $(-0.5466, -0.0131)$.

(c) $\hat{\beta}_0 = 0.77289$, $s_{\hat{\beta}_0} = 0.14534$, $t_{5,01} = 3.365$, so a 98% confidence interval is $0.77289 \pm 3.365(0.14534)$, or $(0.2838, 1.2620)$.

(d) The null and alternate hypotheses are $H_0 : \beta_1 = -0.6$ versus $H_1 : \beta_1 \neq -0.6$.

$$\hat{\beta}_1 = -0.27985, s_{\hat{\beta}_1} = 0.079258, t = (-0.27985 - (-0.6))/0.079258 = 4.039.$$

Since the alternate hypothesis is of the form $\beta_1 \neq b$, the P -value is the sum of the areas to the right of $t = 4.039$ and to the left of $t = -4.039$.

From the t table, $0.002 < P < 0.01$. A computer package gives $P = 0.00993$.

The claim is not plausible.

4. (a) The slope is -0.13468 .

(b) $\hat{\beta}_1 = -0.13468$, $s_{\hat{\beta}_1} = 0.03798$. Since $n = 120$ is large, use the z table to construct a confidence interval.
 $z_{.025} = 1.96$. A 95% confidence interval is $-0.13468 \pm 1.96(0.03798)$, or $(-0.2091, -0.0602)$.

(c) The null and alternate hypotheses are $H_0 : \beta_1 \geq -0.1$ versus $H_1 : \beta_1 < -0.1$.

$$\hat{\beta}_1 = -0.13468, s_{\hat{\beta}_1} = 0.03798. z = (-0.13468 - (-0.1))/0.03798 = -0.91.$$

Since the alternate hypothesis is of the form $\beta_1 < b$, the P -value is the area to the left of $z = -0.91$.

From the z table, $P = 0.1814$.

It is plausible that the slope is greater than or equal to -0.1 .

8. (a) $\bar{x} = 0.840370$, $\bar{y} = 0.826667$, $\sum_{i=1}^n (x_i - \bar{x})^2 = 2.967896$, $\sum_{i=1}^n (y_i - \bar{y})^2 = 2.9446$,
 $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 2.897233$, $n = 27$.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = 0.976191 \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.006304765.$$

The least-squares line is $y = 0.006304765 + 0.976191x$

(b) The null and alternate hypotheses are $H_0 : \beta_0 = 0$ versus $H_1 : \beta_0 \neq 0$.

$$s = \sqrt{\frac{(1 - r^2) \sum_{i=1}^n (y_i - \bar{y})^2}{n - 2}} = 0.0682194,$$

$$\hat{\beta}_0 = 0.006304765, s_{\hat{\beta}_0} = s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} = 0.03577397$$

$t = (0.006304765 - 0)/0.03577397 = 0.1762$. There are $27 - 2 = 25$ degrees of freedom.

Since the alternate hypothesis is of the form $\beta_0 \neq b$, the P -value is the sum of the areas to the left of $t = -0.1762$ and to the right of $t = 0.1762$.

From the t table, $P > 0.80$. A computer package gives $P = 0.8615$.

It is plausible that the intercept is equal to 0.

(c) The null and alternate hypotheses are $H_0 : \beta_1 = 1$ versus $H_1 : \beta_1 \neq 1$.

$$r^2 = \frac{[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})]^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} = 0.9604879, s = \sqrt{\frac{(1 - r^2) \sum_{i=1}^n (y_i - \bar{y})^2}{n - 2}} = 0.0682194,$$

$$\hat{\beta}_1 = 0.976191, s_{\hat{\beta}_1} = \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} = 0.0395989.$$

$t = (0.976191 - 1)/0.0395989 = -0.6013$. There are $27 - 2 = 25$ degrees of freedom.

Since the alternate hypothesis is of the form $\beta_1 \neq b$, the P -value is the sum of the areas to the left of $t = -0.6013$ and to the right of $t = 0.6013$.

From the t table, $0.50 < P < 0.80$. A computer package gives $P = 0.553$.

It is plausible that the slope is equal to 1.

(d) No. It is plausible that $\beta_0 = 0$ and $\beta_1 = 1$, so we cannot conclude that the prediction method is not accurate.

$$(e) \hat{y} = 0.787257, s_{\hat{y}} = s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} = 0.0132258.$$

$t_{25,0.25} = 2.060$. A 95% confidence interval is $0.787257 \pm 2.060(0.0132258)$ or $(0.760, 0.814)$.

(f) Let $y_{0.8} = \beta_0 + \beta_1(0.8)$ denote the mean predicted roughness for steel specimens whose true roughness is $0.8 \mu\text{m}$.

The null and alternate hypotheses are $H_0 : y_{0.8} = 0.75$ versus $H_1 : y_{0.8} \neq 0.75$.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(0.8) = 0.787257, s_{\hat{y}} = s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} = 0.0132258.$$

$t = (0.787257 - 0.75)/0.0132258 = 2.817$. There are $27 - 2 = 25$ degrees of freedom.

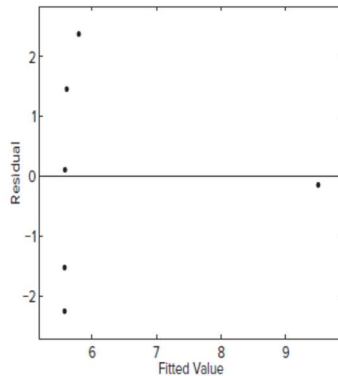
Since the alternate hypothesis is $y_{0.8} \neq 0.75$, the P -value is the sum of the areas to the left of $t = -2.817$ and to the right of $t = 2.817$.

From the t table, $0.002 < P < 0.01$. A computer package gives $P = 0.00933$.

We can conclude that the claim is false.

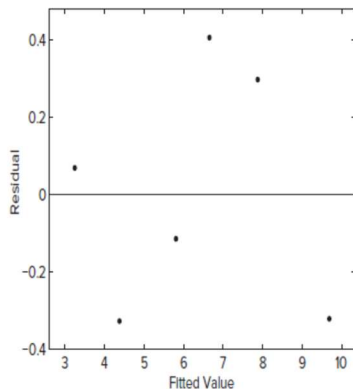
Section 7.4

2. (a)

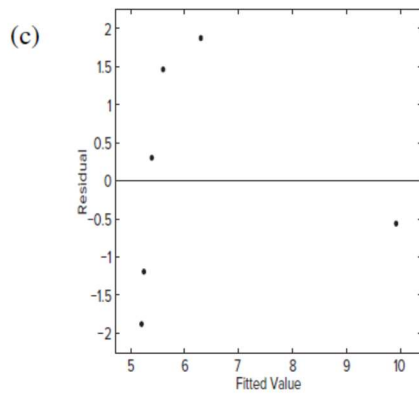


The least-squares line is $y = 5.576 + 0.0142x$. The linear model is not appropriate. The point $(276.02, 9.36)$ is highly influential.

(b)



The least-squares line is $y = 6.148 + 0.629 \ln x$. The linear model appears to be appropriate.

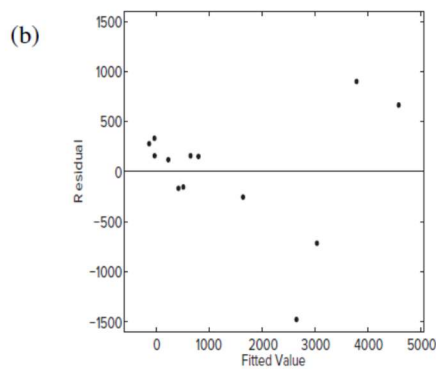


The least-squares line is $y = 5.172 + 0.286x$. The linear model is not appropriate. The point (276.02, 9.36) is highly influential.

(d) The most appropriate model is $y = 6.148 + 0.629 \ln x$. The prediction is $\hat{y} = 6.148 + 0.629 \ln 50 = 8.609$.

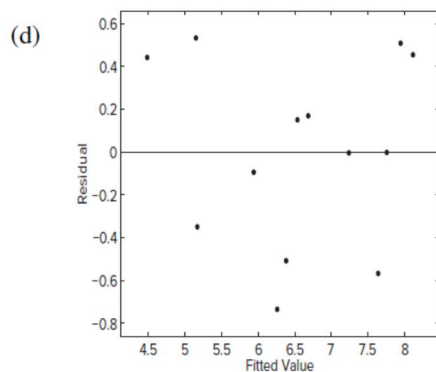
(e) The 95% confidence interval is (8.025, 9.192).

7. (a) $y = -235.32 + 0.695x$.



The residual plot shows a pattern, with positive residuals at the higher and lower fitted values, and negative residuals in the middle. The model is not appropriate.

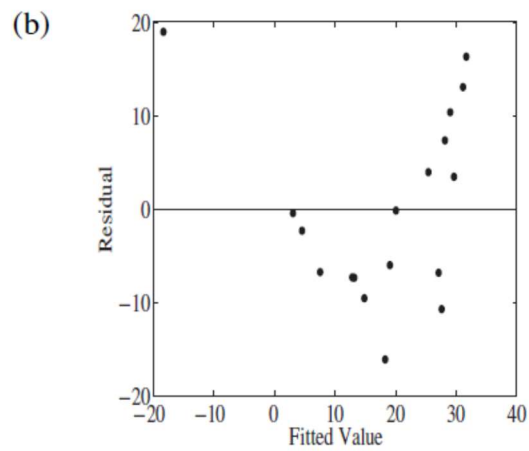
(c) $\ln y = -0.0745 + 0.925 \ln x$.



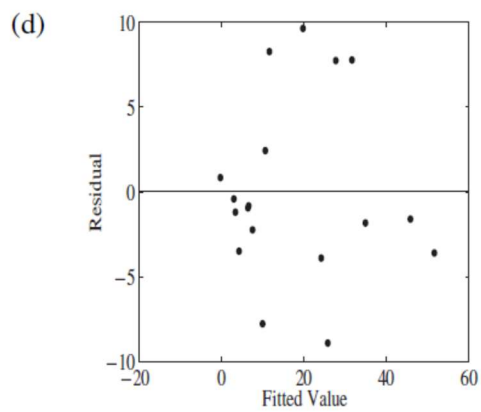
The residual plot shows no obvious pattern. The model is appropriate.

(e) The log model is more appropriate. The 95% prediction interval is (197.26, 1559.76).

10. (a) The least-squares line is $y = 36.632 - 3.2927x$.

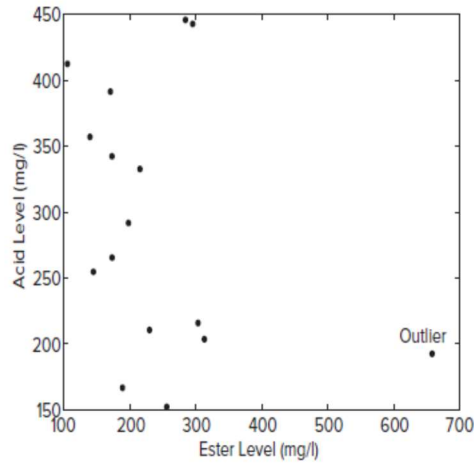


(c) The least-squares line is $y = -5.3657 + 85.439(1/x)$.



(e) The line $y = -5.3657 + 85.439(1/x)$ fits better. The confidence interval is (8.7434, 14.701).

14. (a)



(b) $\hat{\beta}_0 = 352.32$, $s_{\hat{\beta}_0} = 52.774$, $\hat{\beta}_1 = -0.2495$, $s_{\hat{\beta}_1} = 0.1951$.

(c) $t_{14} = -1.28$, $0.20 < P < 0.50$. A computer package gives $P = 0.221$.

(d) $\hat{\beta}_0 = 359.23$, $s_{\hat{\beta}_0} = 91.076$, $\hat{\beta}_1 = -0.2834$, $s_{\hat{\beta}_1} = 0.4108$.

(e) $t_{13} = -0.690$, $P \approx 0.50$. A computer package gives $P = 0.502$.

(f) Not useful. The P -values for the slope are not small, whether or not the outlier is included. Therefore it is plausible that the slope is equal to 0, so a linear model is not helpful for prediction.

Chapter 8

Section 8.1

1. (a) The predicted strength is $26.641 + 3.3201(8.2) - 0.4249(10) = 49.62 \text{ kg/mm}^2$.

(b) By $3.3201(10) = 33.201 \text{ kg/mm}^2$.

(c) By $0.4249(5) = 2.1245 \text{ kg/mm}^2$.

2. (a) $\hat{\beta}_1 = 3.3201$, $s_{\hat{\beta}_1} = 0.33198$, $n = 20$.

There are $20 - 3 = 17$ degrees of freedom. $t_{17,0.025} = 2.110$. A 95% confidence interval is $3.3201 \pm 2.110(0.33198)$, or $(2.620, 4.021)$.

- (b) $\hat{\beta}_2 = -0.4249$, $s_{\hat{\beta}_2} = 0.12606$, $n = 20$.

There are $20 - 3 = 17$ degrees of freedom. $t_{17,0.005} = 2.898$. A 99% confidence interval is $-0.4249 \pm 2.898(0.12606)$, or $(-0.790, -0.060)$.

- (c) $\hat{\beta}_1 = 3.3201$, $s_{\hat{\beta}_1} = 0.33198$, $n = 20$.

The null and alternate hypotheses are $H_0 : \beta_1 \leq 3$ versus $H_1 : \beta_1 > 3$. There are $20 - 3 = 17$ degrees of freedom. $t = (3.3201 - 3)/0.33198 = 0.964$. Since the alternate hypothesis is $\beta_1 > 3$, the P -value is the area to the right of $t = 0.964$.

From the t table, $0.10 < P < 0.25$. A computer package gives $P = 0.174$.

We cannot conclude that $\beta_1 > 3$.

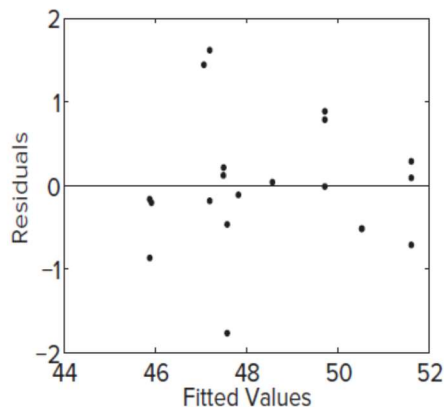
- (d) $\hat{\beta}_2 = -0.4249$, $s_{\hat{\beta}_2} = 0.12606$, $n = 20$.

The null and alternate hypotheses are $H_0 : \beta_2 \geq -0.1$ versus $H_1 : \beta_2 < -0.1$. There are $20 - 3 = 17$ degrees of freedom. $t = [-0.4249 - (-0.1)]/0.12606 = -2.577$. Since the alternate hypothesis is $\beta_2 < -0.1$, the P -value is the area to the left of $t = -2.577$.

From the t table, $0.005 < P < 0.01$. A computer package gives $P = 0.00979$.

We can conclude that $\beta_2 < -0.1$.

3.



The linear model is reasonable. There is no obvious pattern to the plot.

7. (a) $\hat{y} = -0.21947 + 0.779(2.113) - 0.10827(0) + 1.3536(1.4) - 0.0013431(730) = 2.3411$ liters

- (b) By $1.3536(0.05) = 0.06768$ liters

- (c) Nothing is wrong. In theory, the constant estimates FEV_1 for an individual whose values for the other variables are all equal to zero. Since these values are outside the range of the data (e.g., no one has zero height), the constant need not represent a realistic value for an actual person.

8. (a) $\hat{\beta}_1 = 0.779$, $s_{\hat{\beta}_1} = 0.04909$. There are 160 degrees of freedom for error.

Since the number of degrees of freedom is large, use the z table to compute a confidence interval.

$z_{0.025} = 1.96$. A 95% confidence interval is $0.779 \pm 1.96(0.04909)$, or $(0.6828, 0.8752)$.

- (b) $\hat{\beta}_3 = 1.3536$, $s_{\hat{\beta}_3} = 0.2880$. There are 160 degrees of freedom for error.

Since the number of degrees of freedom is large, use the z table to compute a confidence interval.

$z_{0.01} = 2.33$. A 95% confidence interval is $1.3536 \pm 2.33(0.2880)$, or $(0.6826, 2.0246)$.

- (c) $\hat{\beta}_2 = -0.10827$, $s_{\hat{\beta}_2} = 0.0352$.

The null and alternate hypotheses are $H_0 : \beta_2 \geq -0.08$ versus $H_1 : \beta_2 < -0.08$.

There are 160 degrees of freedom for error.

Since the number of degrees of freedom is large, use the z table to find the P -value.

$z = [-0.10827 - (-0.08)]/0.0352 = -0.80$.

Since the alternate hypothesis is of the form $\beta_2 < b$, the P -value is the area to the left of $z = -0.80$. From the z table, $P = 0.2119$. We cannot conclude that $\beta_2 < -0.08$.

- (d) $\hat{\beta}_3 = 1.3536$, $s_{\hat{\beta}_3} = 0.2880$.

The null and alternate hypotheses are $H_0 : \beta_3 \leq 0.5$ versus $H_1 : \beta_3 > 0.5$.

Section 8.2

2. Torque and HP are most nearly collinear. We can tell because their coefficients are significantly different from 0 when either one is in the model without the other, but not significantly different from 0 when both are in the model together.

4. (a)

Predictor	Coef	StDev	T	P
Constant	-1.6415	0.54133	-3.0323	0.013
x_1	3.1415	0.61125	5.1394	0.000

β_0 differs from 0 ($P = 0.013$), β_1 differs from 0 ($P = 0.000$).

(b)

Predictor	Coef	StDev	T	P
Constant	-1.3701	0.57649	-2.3765	0.039
x_2	2.8892	0.6634	4.3552	0.001

β_0 differs from 0 ($P = 0.039$), β_1 differs from 0 ($P = 0.001$).

(c)

Predictor	Coef	StDev	T	P
Constant	-1.8339	0.53319	-3.4395	0.007
x_1	2.0828	0.94564	2.2026	0.055
x_2	1.3006	0.91539	1.4209	0.189

β_0 differs from 0 ($P = 0.007$), the evidence regarding β_1 is borderline ($P = 0.055$), β_2 may not differ from 0 ($P = 0.189$).

- (d) The model $y = \beta_0 + \beta_1 x_1 + \epsilon$. When both x_1 and x_2 are in the model, the coefficient of x_2 is not significantly different from 0. In addition, the value of R^2 is only slightly greater (0.776 vs. 0.725) for the model containing both x_1 and x_2 than for the model containing x_1 alone.

Section 8.3

2. (iii). x_1x_2 , x_1x_3 , and x_2x_3 all have large P -values and thus may not contribute significantly to the fit.

12. (a)

Predictor	Coef	StDev	T	P
Constant	6.6320	1.6806	3.9462	0.002
ln Sc. Dist.	-1.6778	0.36395	-4.6101	0.001
ln Charge	-0.19583	0.20846	-0.9394	0.366

(b)

Predictor	Coef	StDev	T	P
Constant	5.2782	0.86076	6.1320	0.000
ln Sc. Dist.	-1.4652	0.28366	-5.1653	0.000

- (c) The model in part (b) is preferred. In model (a), the coefficient of the variable ln Charge may not be different from 0. It is better to leave this variable out of the model.